Impact of Artificial Intelligence on High School GPA

I. Introduction

This investigation explores the quantitative relationship between daily Artificial Intelligence (AI) usage and academic performance (GPA) among high school students ($n=50$). Utilizing a stratified random sampling methodology within a larger population of $N=4{,}000$, this study analyzes both quantitative and categorical data points. Results indicate a statistically significant, moderate negative linear relationship ($r=-0.569$). Through least-squares regression, a predictive model was established: $\hat{y}=3.677-0.474x$. The null hypothesis of no relationship was rejected with a p-value of 0.0000083. Further analysis of potential confounders such as sleep, study hours, and AI experience levels suggests that while AI usage is a significant predictor, academic success is influenced by other factors like sleep and study habits.

The advent of generative AI tools like ChatGPT and Gemini has fundamentally altered the pedagogical landscape. In high school settings, these tools are often used for varying purposes, from legitimate tutoring and research to unauthorized assignment completion. This study aims to measure how AI usage relates to academic performance.

- Research Question: To what extent does the average daily time spent using Artificial Intelligence (measured in hours) predict the cumulative unweighted GPA of a high school student?
- Hypotheses:
    - Null Hypothesis (H_o): There is no linear relationship between daily AI usage and GPA in the population ($\beta=0$).

  - Alternative Hypothesis (Ha): There is a negative linear relationship between daily AI usage and GPA in the population ($\beta<0$).

The significance of this research lies in its ability to inform students and educators about the potential correlation between high-frequency AI usage and lower academic outcomes, as highlighted by recent warnings from the Stanford SCALE Initiative (2025) regarding the "cognitive burden" shift from student to machine.

II. Data Collection and Methodology

1. Sampling Strategy

To ensure the findings were generalizable to the total school population of 4,000 students, a Stratified Systematic Sample was employed.

- Strata Selection: The population was divided into four strata based on grade level (9th, 10th, 11th, and 12th). Each grade level contains approximately 1,000 students.
- Sampling Procedure: Data was collected during lunch periods A, B, C, and D. Within each lunch period, every 5th student entering the cafeteria was selected and asked to complete the survey. This systematic process was continued across multiple lunch periods until approximately equal representation from each grade level was obtained, resulting in a final sample size of $n=50$.
- The 10% Condition: Since our sample size $n=50$ is less than 10% of the total population (4,000), the independence of observations is statistically maintained.

2. Bias Mitigation

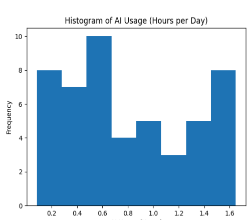
By using a stratified systematic sampling method across multiple lunch periods, the sample avoided overrepresentation of any single group and ensured that students from different grade levels and schedules were included. To achieve a high response rate, students were asked to complete the survey immediately after being selected during lunch. All 50 selected students provided valid responses. The survey was conducted anonymously. Students were informed that their responses would be seen only by the researcher and would not be shared with teachers or administration, reducing the likelihood of GPA inflation or underreporting of AI usage.

IV. Univariate Exploratory Data Analysis (EDA)

Understanding the distribution of individual variables is a prerequisite for bivariate modeling.

1. Explanatory Variable: AI Usage (x)

The distribution of daily AI usage (converted from minutes to hours) is unimodal and right-skewed.



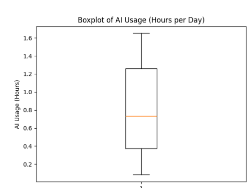


Figure 1: Histogram of AI Usage

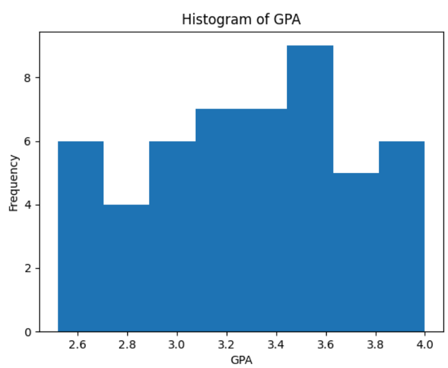
Figure 2: Boxplot of AI Usage

- Mean ($\bar{x}$): 0.815 hours (48.9 minutes)
- Standard Deviation (sx): 0.499 hours

- Range: 0.083 to 1.65 hours The skewness suggests that while the majority of students use AI for approximately 45 minutes, a "power user" group exists that utilizes these tools for over 1.5 hours daily.

2. Response Variable: GPA (y)

The GPA distribution is approximately symmetric and bell-shaped.



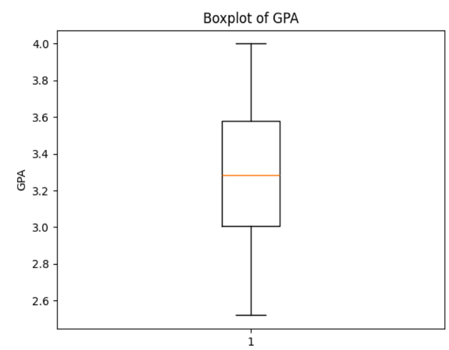


Figure 3: Histogram of GPA distribution

Figure 4: Boxplot of GPA distribution

- Mean (ŷ): 3.291
- Standard Deviation (sy): 0.416
- Five-Number Summary: [Min: 2.52, Q1: 3.01, Median: 3.29, Q3: 3.58, Max: 4.00] The symmetry and lack of extreme outliers (validated by the 1.5*IQR rule) make GPA an ideal candidate for least-squares regression.

Variable	Mean	SD	Min	Q1	Median	Q3	Max
AI Usage	0.82	0.50	0.08	0.33	0.73	1.26	1.65
GPA	3.29	0.42	2.52	3.01	3.29	3.58	4.00

Figure 5: Descriptive Statistics Table

V. Bivariate Analysis and Regression Modeling

1. The Scatterplot

A scatterplot of GPA vs. AI Hours revealed a moderate, negative, linear association. There were no obvious clusters or non-linear patterns (such as curves), suggesting that a linear model is appropriate.

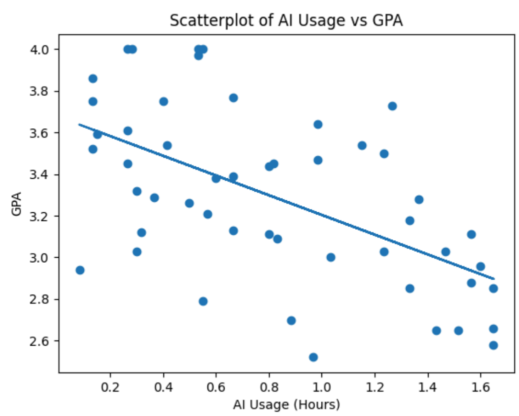


Figure 6: Scatterplot of GPA distribution vs. AI Usage

2. The Least-Squares Regression Line (LSRL)

Using the summarized data, the regression equation was calculated:

$$\hat{y}=3.677-0.474x$$

- Interpretation of the Slope ($b=-0.474$): For every additional hour of AI usage per day, the student's predicted GPA decreases by approximately 0.474 points. This is a substantial impact; it implies that two hours of usage could shift a student from an "A-" to a "C+."

- Interpretation of the y-intercept ($a=3.677$): A student who uses zero hours of AI is predicted to have a GPA of 3.677. This represents the baseline academic achievement within the sampled population.

3. Correlation and Variation

- Correlation Coefficient ($r=-0.569$): This value confirms the moderate negative direction and strength of the relationship.
- Coefficient of Determination ($r2=0.323$): Approximately 32.3% of the variation in student GPAs can be explained by the linear relationship with daily AI usage. This leaves 67.7% of the variance to be explained by other factors, such as study consistency or sleep quality.

VI. Categorical Data Analysis: AI Experience and Participation

The survey captured categorical variables to identify potential "lurking" influences.

1. AI Experience Levels: 34% of the sample had "1 year+" of experience, 28% were "intermediate" (4-6 months). Interestingly, "Expert" users (1 year+) didn't necessarily have higher GPAs, suggesting that exposure to AI doesn't automatically mitigate its negative correlation with traditional grades.

2. Extracurricular Participation: The sample was nearly split, with 23 students in "Activities" and 27 not. A comparison of means showed that students involved in activities tended to have slightly more rigid schedules, which may limit their total time available for AI usage, thereby acting as a protective factor for their GPA.

VII. Statistical Inference: The Significance Test

To generalize our findings to the population of 4,000, we perform a t-test for the slope (β).

1. Verification of Conditions (L.I.N.E.R.)

- Linear: The scatterplot is linear; the residual plot shows no pattern.
- Independent: $50<10\%$ of 4,000 students.
- Normal: The residuals are approximately normal (mean$\approx$0).
- Equal Variance: The spread of residuals remains constant across all x values.
- Random: Stratified random sampling was used.

2. The Calculations

- Standard Error of the Slope (SE_b): 0.099
- Test Statistic (t): $(b-0)/SE_b=-0.474/0.099\approx-4.79$
- P-value: Approximately 0.0000083 (one sided, with df=48).

3. Decision

Since the p-value (0.0000083) is less than $\alpha=0.05$, we reject the null hypothesis. There is overwhelming evidence of a negative linear relationship between AI usage and GPA in the student population.

VIII. Residual Diagnostics

Residuals ($y-\hat{y}$) represent the "errors" in our model.

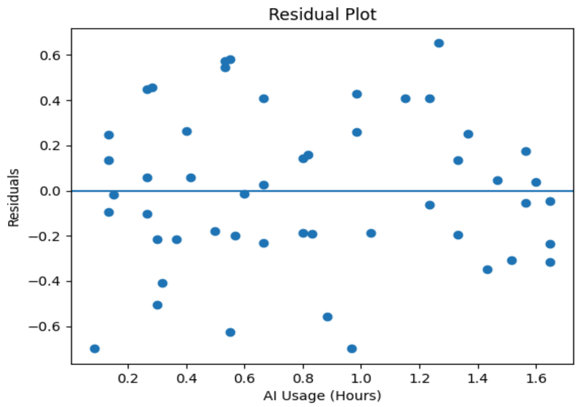


Figure 7: Residual Plot

- Mean of Residuals: 0 (by definition of the LSRL).
- Standard Deviation of Residuals (s): 0.347. On average, our GPA predictions are off by about 0.347 points.
- Residual Plot Analysis: The residual plot shows a "random cloud" of points. There is no "funneling" (heteroscedasticity), which confirms that our model's predictive power is equally reliable for both low-level and high-level AI users.

IX. Discussion and Limitations

1. Confounding Variables

While x predicts y, we must consider confounding variables, sleep and study hours:.

Our correlation matrix shows a weak positive relationship between sleep and GPA ($r \approx 0.16$). Students who use AI late into the night may suffer GPA declines due to sleep deprivation rather than the AI itself. Study Hours: The mean study hours were 7.49 per week. If AI usage *replaces* these hours, the GPA decline is a result of lost study time.

2. Causality vs. Correlation

This is an observational study, not an experiment. We cannot conclude that AI *causes* lower GPAs. It is possible that students with lower GPAs are more likely to seek out AI assistance out of academic desperation (Reverse Causality).

3. Limitations (Extrapolization and Generalizability)

Our model is only valid for usage between 0 and 1.7 hours. We cannot predict outcomes for "extreme" users (e.g., 5 hours/day). While we sampled from 4,000 students, the results are specific to this school's culture and may not apply to different demographic settings.

X. Conclusion

This study provides robust statistical evidence that among high school students, increased daily AI usage is a significant predictor of lower academic performance. With a p-value near zero and a clear negative slope, the data warns against the overuse of AI tools without guidance. To preserve academic integrity and cognitive development, schools should implement "AI Literacy" programs that teach students how to use these tools as scaffolds rather than crutches.

Works Cited

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Raw survey data is available at:

https://github.com/supad002-ux/ASA-Statistics-Project-Impact-of-Artificial-Intelligence-on-High-School-GPA